

Intro to LaTeX

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1 Introduction

We will discuss the usage of LaTeX through Overleaf, which is a great platform for collaboration. We will cover the format of sections in a document, how to write various symbols and expressions in math mode, how to deal with spacing and enumeration, and other miscellaneous items.

2 Document Outline

Everything before `\begin{document}` is called the preamble. This is where we can (optionally) adjust the format of the document, create the header, and import packages to use later. For now, import the packages `graphicx`, `amsmath`, `amssymb`, `amsthm`, and `amsfonts` so that we can use various environments and symbols later.

```
1 \documentclass{article}
2 \setlength{\parindent}{0pt}
3
4 \usepackage[a4paper, lmargin=3cm, rmargin=3cm, tmargin=1cm, bmargin=2cm]{geometry}
5
6 \usepackage{graphicx}
7 \usepackage{amsmath}
8 \usepackage{amssymb}
9 \usepackage{amsthm}
10 \usepackage{amsfonts}
11 \usepackage{tikz}
12 \usepackage{hyperref}
13
14 \title{Intro to LaTeX}
15 \author{Micaela Roth}
16 \date{PROMYS July 2026}
17
18 \begin{document}
19
20 \maketitle
21
22 \section{Introduction}
```

3 Section Title

This is a section, created by the command `\section{Section Title}`.

3.1 Subsection Title

This is a subsection, created by the command `\subsection{Subsection Title}`.

The great thing about organizing a LaTeX document into sections and subsections is that we can easily create a table of contents that updates automatically as we continue to edit the document. Simply insert the command: `\tableofcontents` right after `\maketitle`.

4 Text Formatting

To **bold a section of text**, use `\textbf{bolded text}`

To *italicize a section of text*, use `\textit{italicized text}`

To underline a section of text, use `\underline{underlined text}`

To put “quote a section of text”, use `“quoted text”`

To change the **color of a section of text**, use `\textcolor{red}{colored text}`

5 Math Mode

Ok, let’s get to the reason why we want to use LaTeX in the first place! To go into math mode, surround expressions with dollar signs: `$a+bi \in \mathbb{Z}$`

$$a + bi \in \mathbb{Z}$$

To nicely separate an expression from the other text that appears in the document, use display mode by surrounding expressions with pairs of dollar signs: `$$a+bi \in \mathbb{Z}$$`

$$a + bi \in \mathbb{Z}$$

To put multiple lines of text into math mode, we can wrap the text with `\[` and `\]`.

It can also be useful to insert text while in math mode. We can do this using the `\text` command: `$$\text{Consider } a+bi \text{ as an element of } \mathbb{Z}$$`

Consider $a + bi$ as an element of $\mathbb{Z}$

Let’s continue to do some more examples.

$$\frac{3}{5} \rightarrow \frac{3}{5}$$

$$\sqrt{2} \rightarrow \sqrt{2}$$

$$\sqrt[3]{2} \rightarrow \sqrt[3]{2}$$

$$e^{\pi i} \rightarrow e^{\pi i}$$

$$\mathbb{Z}_p \rightarrow \mathbb{Z}_p$$

In math mode, we often use braces to group a block of text together. To “escape” LaTeX misinterpreting braces (or other symbols) as a grouping indicator, use a backslash: `\{1,2,3,4,5\} \rightarrow \{1,2,3,4,5\}`.

The backslash to escape can also be used to create spaces in math mode: `a^2 \ + \ b^2 \ = \ c^2`
 $\rightarrow a^2 + b^2 = c^2$.

5.1 Math Symbols

<code>\in</code>	$\in$
<code>\notin</code> or <code>\not\in</code>	$\notin$
<code>\le</code>	$\leq$
<code>\ge</code>	$\geq$
<code>\ne</code>	$\neq$
<code>\subset</code>	$\subset$
<code>\subseteq</code>	$\subseteq$
<code>\not\subset</code>	$\not\subset$
<code>\sim</code>	$\sim$
<code>\equiv</code>	$\equiv$
<code>\cong</code>	$\cong$

<code>\forall</code>	$\forall$
<code>\exists</code>	$\exists$
<code>\rightarrow</code>	$\rightarrow$
<code>\leftarrow</code>	$\leftarrow$
<code>\implies</code>	$\implies$
<code>\impliedby</code>	$\impliedby$
<code>\iff</code>	$\iff$
<code>\mapsto</code>	$\mapsto$
<code>\times</code>	$\times$
<code>\cdot</code>	$\cdot$
<code>\circ</code>	$\circ$
<code>\Box</code>	$\square$

<code>\cap</code>	$\cap$
<code>\bigcap</code>	$\bigcap$
<code>\cup</code>	$\cup$
<code>\bigcup</code>	$\bigcup$
<code>\sum</code>	$\sum$
<code>\prod</code>	$\prod$
<code>\int</code>	$\int$
<code>\lfloor</code>	$\lfloor$
<code>\rfloor</code>	$\rfloor$
<code>\lceil</code>	$\lceil$
<code>\rceil</code>	$\rceil$

<code>\alpha</code>	α
<code>\beta</code>	β
<code>\gamma</code>	γ
<code>\delta</code>	δ

We can customize some of these symbols. We can label arrows on either the top, bottom, or both: `\xrightarrow[xyz]{abc}`

$$\xrightarrow[xyz]{abc}$$

For taking the sum, product, union, etc. over a set: `\sum_{i=1}^n a_n`

$$\sum_{i=1}^n a_n$$

When these “taller” expressions are in display mode, they don’t play very nicely with other symbols such as parentheses:

$$(\sum_{i=1}^n (\sum_{j=1}^m A_{ij}))$$

To fix this, use `\left` or `\right` before each of the parentheses respectively:

$$\left(\sum_{i=1}^n \left(\sum_{j=1}^m A_{ij} \right) \right)$$

5.2 Math Fonts

The most common used fonts in math mode for variables are `mathbb` and `mathcal`. Examples:

$$\mathbb{Z} \rightarrow \mathbb{Z}$$

$$\mathbb{Q} \rightarrow \mathbb{Q}$$

$$\mathbb{R} \rightarrow \mathbb{R}$$

$$\mathcal{A} \rightarrow \mathcal{A}$$

$$\mathcal{B} \rightarrow \mathcal{B}$$

$$\mathcal{C} \rightarrow \mathcal{C}$$

This is a lot to type out each time, so we can also create some text shortcuts in the preamble:

```
\newcommand{\Z}{\mathbb{Z}}
```

This allows us to type `\Z` instead of `\mathbb{Z}` every time we want a $\mathbb{Z}$ in math mode.

6 Spacing, Enumeration, Tables

6.1 Spacing

Spacing is important for both organization and readability, especially in documents with many equations. To insert vertical spaces anywhere, use the commands: `\smallskip`, `\medskip` or `\bigskip`, followed by a new line.

To get to a new line within math mode, use `\\`. To start a new page, use `\newpage`.

To keep track of equations, we can enumerate them using the `equation` environment (from `amsmath`). Surround equations with `\begin{equation}` and `\end{equation}`.

$$a^2 + b^2 = c^2 \tag{1}$$

$$P_k = a_k P_{k-1} + P_{k-2} \tag{2}$$

The `equation` environment automatically centers equations. We can also center any block of text by surrounding it with `\begin{center}` and `\end{center}`.

For longer sequences of equalities, we can use the `align` environment (from `amsmath`) by putting `&` right after each new line `\\` to make it start directly below the last `&`. It typically looks nice to align lines based on where the first equals sign is, so this will look like: `\\ &=`.

$$\begin{aligned}\sqrt{67} &= 8 + \frac{1}{\frac{1}{\sqrt{67}-8}} = 8 + \frac{1}{\frac{\sqrt{67}+8}{67-64}} = 8 + \frac{1}{\frac{\sqrt{67}+8}{3}} = 8 + \frac{1}{5 + \frac{\sqrt{67}+8-15}{3}} \\ &= 8 + \frac{1}{5 + \frac{1}{\frac{3}{\sqrt{67}-7}}} = 8 + \frac{1}{5 + \frac{1}{\frac{3(\sqrt{67}+7)}{67-49}}} = 8 + \frac{1}{5 + \frac{1}{\frac{\sqrt{67}+7}{6}}} = \dots\end{aligned}$$

(Note: the above equation is written in the `align*` environment to avoid equation enumeration).

6.2 Enumeration

To create a numbered list, we use the built-in `enumerate` environment.

```
\begin{enumerate}
  \item ...
  \item ...
\end{enumerate}
```

For example:

1. This is
2. a list!

We can also choose how the numbers (or not-numbers) before each item appear, using `\item[number]` For example, if we wanted to confuse our readers, we could write:

```
\begin{enumerate}
  \item[(3)] Firstly
  \item[(1)] Secondly
  \item[(2)] Thirdly
  \item[(a)] 67
\end{enumerate}
```

This results in:

- (3) Firstly
- (1) Secondly
- (2) Thirdly
- (a) 67

Another helpful type of list is a bulleted list, which we can create using the `itemize` environment.

```
\begin{itemize}
  \item ...
  \item ...
\end{itemize}
```

For example:

- Here is a
- bulleted list!

6.3 Tables, Matrices

To create tables, we use the `tabular` environment, surrounding the text in the table by `\begin{tabular}` and `\end{tabular}`.

The expression with *c*'s and vertical bars that appears after `\begin{tabular}` controls the vertical line format between columns in the table. Elements are separated into cells using `&`, new rows are indicated by `\\`, and horizontal lines between rows are inserted using `\hline`.

```
\begin{tabular}{c|c|c}
\hline
a & b & c \\
\hline
1 & 2 & 3 \\
\hline
\end{tabular}
```

a	b	c
1	2	3

Matrices take on a similar format as tables. We can create matrices with either parentheses or brackets, which use the environments `pmatrix` and `bmatrix` (from `amsmath`) respectively. For example:

```
\begin{bmatrix}
a & b \\
c & d
\end{bmatrix}
```

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

One thing to note is that LaTeX expects matrices to be written in made mode. To do this, we can use the multi-line form of math mode and wrap the text with `\[` and `\]`.

7 Useful Packages

External packages can be useful for formatting theorems and proofs, creating figures, and typing math symbols. We import packages in the preamble, for example using the command `\usepackage{amsthm}`.

7.1 amsthm

After including the package, we can also set up some shortcut commands in the preamble. Notice that we can set up shortcuts for different words like `lem` for `Lemma`, but we typically still use the `[thm]` format.

```
\theoremstyle{plain}
\newtheorem{thm}{Theorem}
\newtheorem{cor}[thm]{Corollary}
\newtheorem{lem}[thm]{Lemma}
\newtheorem{defn}[thm]{Definition}
\newtheorem{prop}[thm]{Proposition}
\newtheorem{ex}[thm]{Example}
\newtheorem{rmk}[thm]{Remark}
```

This allows us to format things like theorems, corollaries, lemmas, and proofs nicely. Theorems can also be labeled so that they can be cited later. For example:

```
\begin{thm} \rm \label{fav-theorem}
Theorem statement.
\end{thm}
```

(Note: the `\rm` command after `\begin{theorem}` keeps the theorem statement from being automatically italicized).

Theorem 1. Theorem statement.

```
\begin{proof}
  This proof is left as an exercise to the reader.
\end{proof}
```

Proof. This proof is left as an exercise to the reader. □

And the citation:

My favorite theorem is Theorem `\ref{fav-theorem}`.

My favorite theorem is Theorem 1.

7.2 graphicx

Graphicx allows insertion of images into documents. After including the package and uploading an image into the file tree, we can directly insert the image (and adjust the scale if necessary) using the following command:

```
\includegraphics[scale=0.5]{filename.png}
```

8 Figures

The `figure` environment, similarly to the theorem organization methods from `amsthm`, can be used to enumerate, label, and caption figures in a document.

After creating a figure in TikZ (like the one below) or through other means, wrap all of the text in `\begin{figure}` and `\end{figure}`. This automatically enumerates the figures in your document.

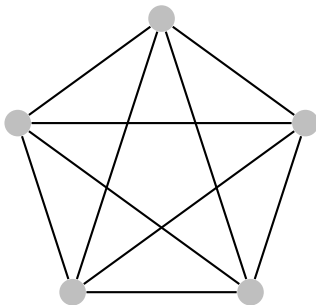


Figure 1: The complete graph K_5 .

To write a caption, use the `\caption` command after the figure code but before `\end{figure}`.

We can also label and reference figures just like we did for theorems. However, an important thing to note is that the `\label` command should go right *after* the caption, so that the end of the figure environment looks like this:

```
...
\caption{The complete graph  $K_5$ .}
\label{K5-graph}
\end{figure}
```

9 Other Features

- Bibliography: www.overleaf.com/learn/latex/Bibliography_management_with_bibtex
- Beamer presentation: www.overleaf.com/learn/latex/Beamer
- TikZ (manually creating figures): www.overleaf.com/learn/latex/TikZ_package
- Quiver (figure construction $\rightarrow$ LaTeX code): q.uiver.app
- Detexify (written work $\rightarrow$ LaTeX code): detexify.kirelabs.org

10 References

For guidance on how to use other features, visit this documentation site: www.overleaf.com/learn